

Network Science

Lecture 06: Structural Balance

5942UE Network Science

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Structural Balance

We now study **signed networks** where relations can be positive or negative.

Focus

- complete signed graphs
- structural balance and its global theorem
- applications to politics and online trust
- weak and relaxed balance

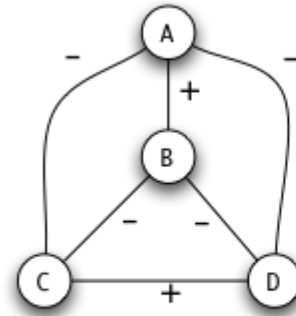
Module 1

Structural Balance Theorem

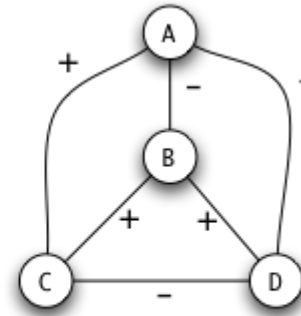
Guiding Question

What global network structures are compatible with psychologically plausible signed triangles?

Structural Balance Property



balanced



not balanced

A complete signed graph is **structurally balanced** if every triangle is either all-positive or has exactly one positive edge.

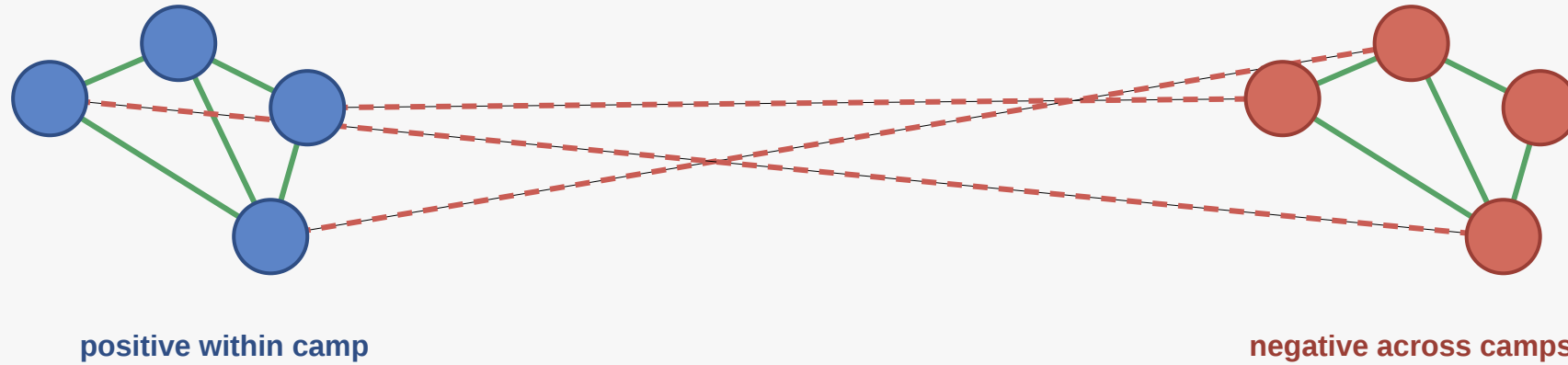
Core Result

If a complete signed graph is balanced, then its nodes can be partitioned into:

- one all-positive block
- or two internally positive blocks with all negative edges between them

Balanced as Camps

Balanced signed structure collapses into camps



Interpretation

If the graph is balanced, it must collapse into either:

- one all-friends block
- or two internally friendly blocks with hostility between them

This is a classic example of a local constraint producing strong global organization.

Module 2

Applications

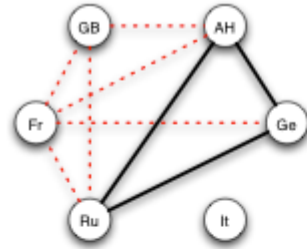
Guiding Question

What do signed-network models help us explain in real settings?

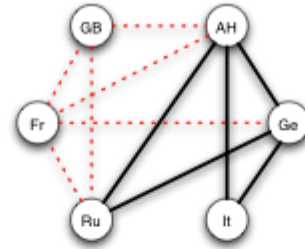
Formal Intuition

Signed networks extend graphs by giving each edge a **positive or negative label**.

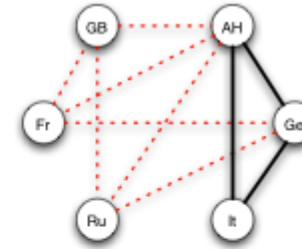
International Relations



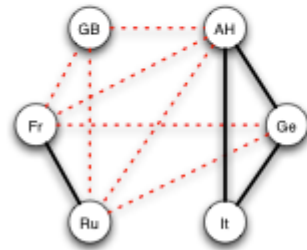
(a) *Three Emperors' League 1872-81*



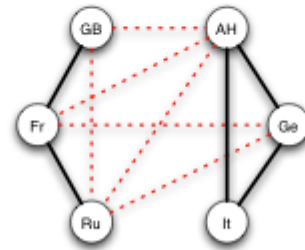
(b) *Triple Alliance 1882*



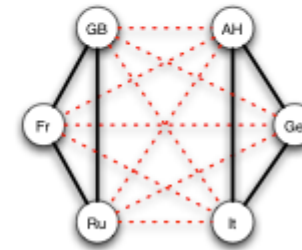
(c) *German-Russian Lapse 1890*



(d) *French-Russian Alliance 1891-94*



(e) *Entente Cordiale 1904*



(f) *British Russian Alliance 1907*

Structural balance can describe alliance formation and polarization into camps.

Online Trust

Signed ties can also model:

- **trust and distrust**
- **recommendation and warning**
- **support and opposition**

Interpretation

On trust platforms, positive and negative links do not behave exactly like undirected friendship and hostility, but the balance framework still suggests useful questions:

- when should distrust propagate?
- when do trust triangles stabilize?
- what patterns indicate factional structure?

Applications are strongest when the signed model is treated as an approximation, not as a literal truth

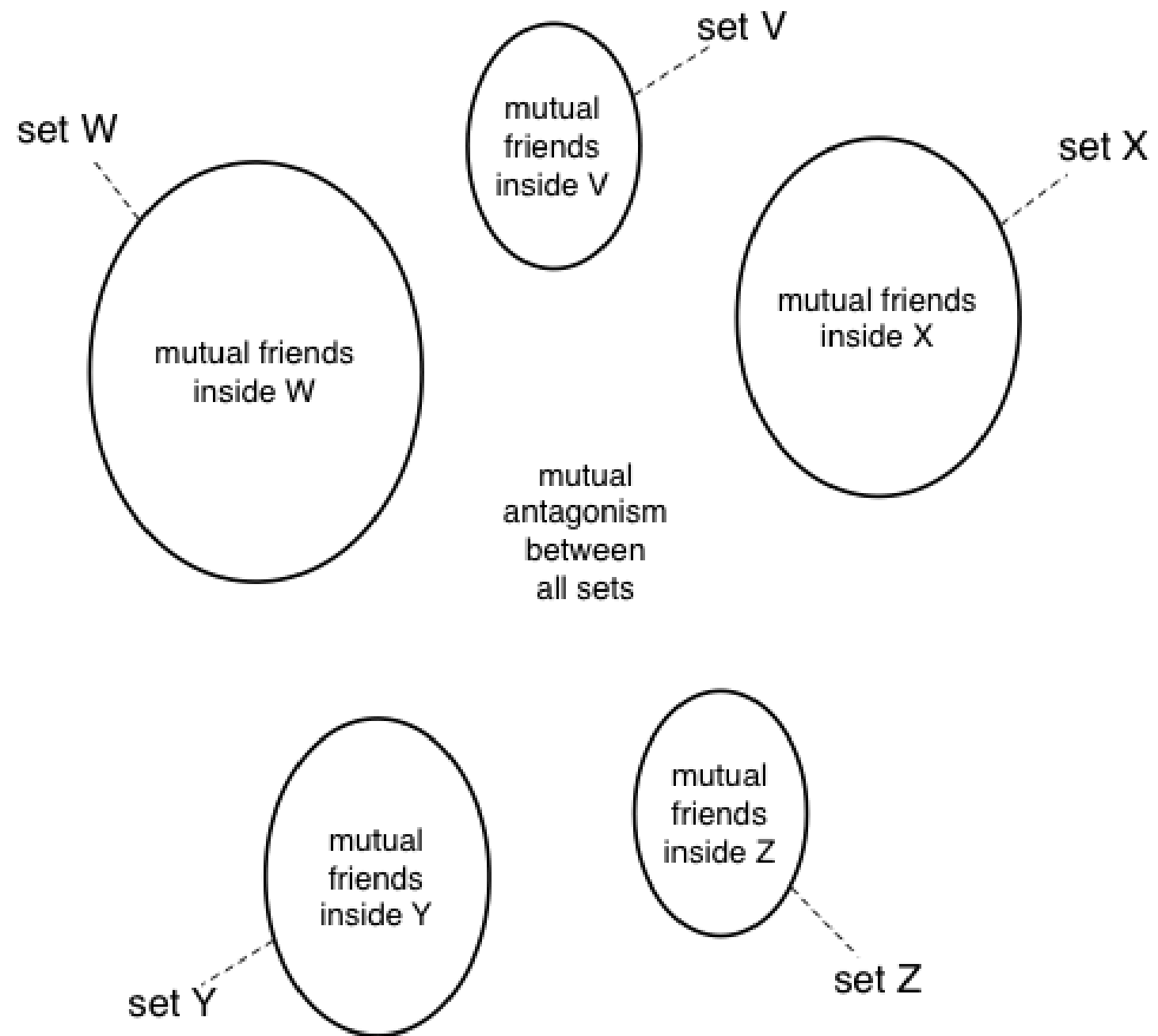
Module 3

Weak Structural Balance

Guiding Question

What changes if "the enemy of my enemy" is not forced to become my friend?

Counterexample



Weak Balance

Weak structural balance forbids triangles with exactly two positive edges and one negative edge, but allows all-negative triangles.

Interpretation

That relaxation changes the global theorem:

- instead of exactly two camps
- the graph may split into **multiple** internally positive groups
- with negative ties between groups

Changing one local rule can change the entire global picture.

Module 4

Relaxed and Incomplete Balance

Guiding Question

How can we reason about balance when graphs are incomplete or only approximately balanced?

Incomplete Graphs

Not every pair of actors has a known signed relation. That forces us to ask whether balance can still be inferred from **partial evidence**.

Odd Negative Cycles

odd number of negative edges on a cycle $\Rightarrow$ not balanced

Formal Intuition

Once completeness is dropped, balance is no longer a simple triangle-only condition.

Instead, the question becomes whether the observed graph is compatible with an **approximately balanced partition**.

Approximate Balance

In real data, a graph can be **mostly balanced** without satisfying the theorem perfectly. Then the right question is whether it is close to a balanced partition.

Relaxed balance is more realistic, but it replaces sharp theorems with softer structural diagnostics.

Wrap-Up

- local triangle constraints imply strong global structure
- signed networks are analytically powerful but assumption-heavy
- relaxing the assumptions changes both the theorem and the interpretation